The *E. coli* Datasets, and the Short List That Can Benchmark a Gene-Sequence Encoding

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The dev tree serves 68 *Escherichia coli* datasets holding 4,561,865 records, built as LMDBs from their own releases and read here from the snapshot of 2026-10-21 (Table 2). Every column of that table is read from a built store; nothing in it is typed by hand.

A **gene-sequence encoding** is one vector per coding sequence and nothing else: no cell graph, no annotation, no gene identity beyond what the sequence carries. GenSLM is the instance at hand, a codon language model pretrained on prokaryotic genes, and `torchcell/datasets/genslm.py` already builds such a store for any genome whose genes expose a coding sequence. Benchmarking that encoding means asking, dataset by dataset, whether the vector of a gene the model has not been fit on predicts that gene's label. Of the 68 datasets, 35 attach a label to one gene, one gene pair or one inserted construct (Table 1); far fewer carry enough labeled genes, on a task worth scoring, to make a benchmark. Sec. 2 sets the rule and gives the short list (Table 3): 4 core datasets, one per task family, each with between 2,359 and 4,181 labeled genes, and a second tier of the same size. Sec. 3 names what a score on any of them does not show.

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1 The *E. coli* datasets ✗

Table 2 lists the 68 datasets in 13 phenotype families. **Genotypes** and **Env** are the distinct stored genotype and environment payloads, **Instances** the record count, and **Signal** the gzip size in bits of the concatenated stored instances, the same relative proxy the yeast table uses. A dataset whose strains are all the one wild type, such as a proteome panel across growth conditions, shows one genotype and as many environments as conditions; a deletion screen shows one genotype per strain.

Two columns are specific to this document. The **sequence-only fit** tier says how well a model that sees only sequence embeddings fits the dataset, and the **label unit** names what one label attaches to once the cell graph is dropped. Both are derived by the probe script from where the label sits in the cell graph (a whole-strain readout, a node, a gene pair, a metabolite) and from the modal number of edits in a sample of 400 genotypes per store. The tiers are structural: *best* when the label sits on one gene, one gene pair or triple, or one inserted construct, so that a per-gene or per-pair representation is the whole input; *good* when the label needs the readout entity crossed with the perturbed genes or with the condition, which a pair of vectors still carries; *weak* for a multi-edit strain where only its heterologous genes have a sequence to stand on; *no* for a strain readout with no single edit to attach a sequence to. Table 1 sums the datasets and instances per tier. A tier is a reading of the genotype, not a model result; it says what a sequence-only model could be given, not what it would score.

Table 1: *E. coli* datasets and instances per sequence-only fit tier, as `sequence_fit` in the probe script derives it from where the label sits and the modal edit count of the genotype sample.

Sequence-only fit	Datasets	Instances	What the tier means
best	35	639,394	the label sits on one gene, a gene pair or triple, or one inserted construct
good	29	3,921,964	readout entity x perturbed gene(s), or gene x condition
weak	1	72	a multi-edit strain, only its heterologous genes carry a sequence
no	3	435	a natural genome, or a whole-strain readout with no single edit
Total	**68**	**4,561,865**	

Table 2: The *E. coli* datasets served from the dev tree, snapshot 2026-10-21, grouped by phenotype family. *Genotypes* and *Env* count distinct stored genotype and environment payloads; *Instances* is the record count; *Signal* the gzip size in bits of the concatenated stored instances (perturbation, environment and phenotype). *Seq. fit* is the sequence-only fit tier of Table 1 and *Label unit* what one label attaches to without the cell graph, both read from the genotype of 400 evenly spaced records per store.

Dataset	Genotypes	Env	Instances	Phenotype	Shape	Role	Signal (bits)	Seq. fit	Label unit
Environmental / chemogenomic response									
CrisprPineneToleranceNiu2019	54	1	54	environment_response (log2_ratio)	scalar	global	4.1×10^4	best	1 gene (1-6 edits)
CrispriChemgenChoe2025	39,291	12	470,502	environment_response (log2_ratio)	scalar	global	2.4×10^8	good	1 gene x condition
CrispriGeneFitnessWang2018	13,641	5	20,158	environment_response (log2_ratio)	scalar	global	1.5×10^7	good	1 gene x condition (1-7 edits)
CrispriGuideFfaEnrichmentFang2025	15,711	1	15,711	environment_response (log2_ratio)	scalar	global	7.3×10^6	best	1 gene (1-5 edits)
CrispriGuideFitnessWang2018	53,203	5	240,481	environment_response (log2_ratio)	scalar	global	1.6×10^8	good	1 gene x condition
CrispriKnockdownCui2018	71,696	1	143,392	environment_response (log2_ratio)	scalar	global	7.3×10^7	best	1 gene (1-7 edits)
CrispriScreenRousset2018	40,592	4	92,252	environment_response (log2_ratio)	scalar	global	4.2×10^7	good	1 gene x condition
DoublingTimeCaglar2017	1	16	55	environment_response (growth_rate)	scalar	global	3.2×10^4	no	unperturbed strain readout x condition
EnvChemgenGirgis2009	3,976	17	66,287	environment_response (z_score)	scalar	global	3.7×10^7	good	1 gene x condition
EnvChemgenNichols2011	3,967	324	1,286,962	environment_response (z_score)	scalar	global	4.1×10^8	good	1 gene x condition
EnvChemgenShiver2016	3,846	51	210,998	environment_response (z_score)	scalar	global	9.9×10^7	good	1 gene x condition
EnvChemgenWang2015	46	1	46	environment_response (log2_ratio)	scalar	global	3.0×10^4	best	1 gene
EnvChemgenWang2024	4,419	6	26,514	environment_response (log2_ratio)	scalar	global	9.3×10^6	good	1 gene x condition
FitnessBrowserEcoli	3,789	3	22,734	environment_response (log2_ratio)	scalar	global	1.1×10^7	good	1 gene x condition
GrowthRateLamoureux2023	21	54	354	environment_response (growth_rate)	scalar	global	2.1×10^5	no	unperturbed strain readout x condition (0-1 edits)
GrowthRateS23Schmidt2016	1	22	26	environment_response (growth_rate)	scalar	global	2.0×10^4	no	unperturbed strain readout x condition
GrowthYieldBaba2006	3,912	3	11,735	environment_response (optical_density)	scalar	global	3.3×10^6	good	1 gene x condition
MismatchCrispriFitnessHawkins2020	24,186	1	24,186	environment_response (relative_growth_rate)	scalar	global	1.4×10^7	best	1 gene
PhageRbTnseqMutalik2020	3,716	71	287,815	environment_response (log2_ratio)	scalar	global	2.1×10^8	good	1 gene x condition
RbTnseqPrice2018Ecoli	3,789	104	598,662	environment_response (log2_ratio)	scalar	global	4.0×10^8	good	1 gene x condition
RbTnseqTeteneva2024	3,691	18	66,162	environment_response (log2_ratio)	scalar	global	2.7×10^7	good	1 gene x condition
RbTnseqWetmore2015	3,646	85	353,662	environment_response (log2_ratio)	scalar	global	2.4×10^8	good	1 gene x condition
Expression (RNA-seq)									
MrnaFractionBalakrishnan2022	3	14	28	mrna_number_fraction (rnaseq_mrna_number_fraction)	vector (4342)	node	7.2×10^6	best	1 gene, unperturbed strain x condition (0-2 edits)
RnaseqCaglar2017	1	57	152	expression_tpm (rnaseq_tpm)	vector (4196)	node	6.7×10^7	best	1 gene, unperturbed strain x condition
RnaseqGuzman2018	7	2	14	expression_tpm (rnaseq_tpm)	vector (4329)	node	4.5×10^6	good	readout gene x 3 perturbed gene(s) x condition (0-5 edits)
RnaseqLamoureux2023	74	119	1,035	expression_tpm (rnaseq_tpm)	vector (4257)	node	5.8×10^8	best	1 gene, unperturbed strain x condition (0-3 edits)

Table 2, continued

Dataset	Genotypes	Env	Instances	Phenotype	Shape	Role	Signal (bits)	Seq. fit	Label unit
RnaseqPublicK12Lamoureux2023	72	198	1,675	expression_tpm (rnaseq_tpm_from_released_counts)	vector (4355)	node	9.5×10^8	best	1 gene, unperturbed strain x condition (0-4 edits)
Fitness									
CarbonSourceTong2020	3,714	30	111,420	fitness	scalar	global	4.8×10^7	good	1 gene x condition
CrispriCrossRachwalski2024	14,209	15	48,900	fitness	scalar	global	1.9×10^7	good	gene pair, strain readout x condition (1-2 edits)
GrowthAucRapp2026	1,515	1	1,515	fitness	scalar	global	7.6×10^5	best	1 gene
GrowthRateCampos2018	3,664	1	3,664	fitness	scalar	global	1.1×10^6	best	1 gene
GrowthRateChoe2019	6	1	8	fitness	scalar	global	1.1×10^4	best	1 gene (0-21 edits)
GrowthRateSchmidt2016	3	2	6	fitness	scalar	global	8.5×10^3	good	1 gene x condition
GrowthWang2015	46	1	46	fitness	scalar	global	2.2×10^4	best	1 gene
TranscriptionFactorKnockoutChoe2019	2	1	2	fitness	scalar	global	5.8×10^3	best	1 gene
Flux									
FluxIshii2007	25	5	28	net_flux (c13_mfa_net_flux_percent_of_specific_glucose_uptake)	vector (42)	bipartite node	6.0×10^4	good	reaction x 1 perturbed gene(s) x condition (0-1 edits)
FluxLi2021	3	1	3	net_flux (c13_mfa_net_flux_mmol_per_gdw_per_h)	vector (99)	bipartite node	5.7×10^4	good	reaction x 1 perturbed gene(s)
Gene interaction									
GeneInteractionBabu2014	42,703	1	42,705	gene_interaction	scalar	edge	1.2×10^7	best	gene pair
GeneInteractionButland2008	314,769	1	314,769	gene_interaction	scalar	edge	8.2×10^7	best	gene pair
Metabolite									
ExometaboliteBrunk2016	9	28	126	metabolite_level (hplc_extracellular_concentration_g_per_l)	vector (6)	bipartite node	2.3×10^5	good	metabolite x 9 perturbed gene(s) x condition (0-10 edits)
MetaboliteIntensityRapp2026	407	1	407	metabolite_level (fi_ms_annotated_feature_absolute_intensity_mean_of_2_plates)	vector (2)	bipartite node	3.9×10^5	good	metabolite x 1 perturbed gene(s)
MetabolomeBrunk2016	9	26	117	metabolite_level (lc_ms_concentration_um)	vector (51)	bipartite node	4.5×10^5	good	metabolite x 9 perturbed gene(s) x condition (0-10 edits)
MetabolomeFuhrer2017	3,806	1	3,806	metabolite_level (fia_tof_ms_ion_modified_z_score)	vector (7534)	bipartite node	1.9×10^9	good	metabolite x 1 perturbed gene(s)
MetabolomeIshii2007	25	5	28	metabolite_level (ce_tofms_intracellular_concentration_mm)	vector (130)	bipartite node	4.9×10^5	good	metabolite x 1 perturbed gene(s) x condition (0-1 edits)
MetabolomeRapp2026	1,497	1	1,497	metabolite_level (fi_ms_iml1515_feature_fold_change_vs_batch_median_mean_of_2_plates)	vector (1321)	bipartite node	5.6×10^8	good	metabolite x 1 perturbed gene(s)
MetabolomeSchastnaya2021	103	5	200	metabolite_level (fia_tof_ms_ion_log2_fold_change_vs_wild_type)	vector (284)	bipartite node	3.6×10^6	good	metabolite x 1 perturbed gene(s) x condition
TargetedMetabolomeRapp2026	407	1	407	metabolite_level (lc_msms_eic_peak_height_fold_change_vs_library_median)	vector (2)	bipartite node	3.6×10^5	good	metabolite x 1 perturbed gene(s)
Morphology									
MorphologyCampos2018	3,664	1	3,664	morphology	vector (15)	global	8.5×10^6	best	1 gene
Phosphoproteome									

Table 2, continued

Dataset	Genotypes	Env	Instances	Phenotype	Shape	Role	Signal (bits)	Seq. fit	Label unit
PhosphositeIntensityPotel2018	1	2	12	phosphosite_intensity (fe3_imac_lc_msms_ maxquant_summed_xic)	vector (989)	node	1.3×10^7	best	1 gene, unperturbed strain x condition
Product titer									
BiofuelTiterBrunk2016	8	18	72	titer	scalar	global	1.5×10^5	weak	set of 9 edits incl. heterologous genes (7-10 edits)
IsopentenolTiterFoo2014	9	1	9	titer	scalar	global	1.7×10^4	best	1 heterologous construct
PineneFoldChangeNiu2019	40	1	40	titer	scalar	global	2.3×10^4	best	1 gene (1-6 edits)
Promoter activity									
PromoterReporterMohiuddin2022	1,809	27	69,480	promoter_activity	scalar	node	3.3×10^7	best	inserted DNA (promoter or reporter) x condition
Protein abundance									
ProteinFoldChangeCaglar2017	1	8	8	protein_fold_change (deseq2_wald_log2_fold_ change_design_batch_ plus_condition)	vector (4180)	node	1.1×10^7	best	1 gene, unperturbed strain x condition
ProteomeBrunk2016	9	18	81	protein_abundance (srm_ protein_peak_area)	vector (68)	node	4.2×10^5	good	readout gene x 9 perturbed gene(s) x condition (0-10 edits)
ProteomeCaglar2017	1	40	105	protein_abundance (lcmsms_spectral_count_ deseq2_size_factor_ normalized)	vector (4196)	node	2.5×10^7	best	1 gene, unperturbed strain x condition
ProteomeIshii2007	25	5	28	protein_abundance (lc_ ms_ms_absolute_mg_ protein_per_g_dry_cell_ weight)	vector (57)	node	2.9×10^5	best	1 gene, its own label x condition (0-1 edits)
ProteomeMori2021	1	1	7	protein_abundance (absolute_protein_mass_ fraction_xtop_dia_swath_ riboprofiling_scaled)	vector (1897)	node	1.1×10^6	best	1 gene, unperturbed strain
ProteomeSchmidt2016	1	21	21	protein_abundance (absolute_protein_copies_ per_cell_label_free_sid_ anchored)	vector (2359)	node	1.1×10^7	best	1 gene, unperturbed strain x condition
ProteomeSrmSet1Schmidt2016	1	18	18	protein_abundance (absolute_protein_copies_ per_cell_srm_sid_ dataset1_technical_sd)	vector (41)	node	1.5×10^5	best	1 gene, unperturbed strain x condition
ProteomeSrmSet2Schmidt2016	1	21	21	protein_abundance (absolute_protein_copies_ per_cell_srm_sid_ dataset2_biological_sd)	vector (31)	node	1.3×10^5	best	1 gene, unperturbed strain x condition
Protein synthesis / turnover									
ProteinSynthesisRateLi2014	1	3	3	synthesis_rate (ribosome_ profiling_footprint_ density_absolute_ synthesis_rate)	vector (3041)	node	3.7×10^5	best	1 gene, unperturbed strain x condition
ProteinTurnoverGupta2024	6	8	13	degradation_rate (n15_ ammonium_tmtproc_ total_turnover_rate_per_ hour)	vector (2553)	node	1.1×10^7	best	1 gene, unperturbed strain x condition (0-3 edits)
Viability									
GeneEssentialityBaba2006	4,288	1	4,288	is_essential	scalar	node	1.8×10^6	best	1 gene, its own label
GeneEssentialityGerdes2003	3,739	1	3,739	is_essential	scalar	node	2.0×10^6	best	1 gene, its own label
GeneEssentialityGoodall2018	8,309	2	8,309	is_essential	scalar	node	5.2×10^6	best	1 gene, its own label x condition
GeneEssentialityPrice2018Ecoli	324	1	324	is_essential	scalar	node	2.6×10^5	best	1 gene, its own label

Table 2, continued

Dataset	Genotypes	Env	Instances	Phenotype	Shape	Role	Signal (bits)	Seq. fit	Label unit
GradedGeneEssentialityGoodall2018	317	2	317	essentiality_call	scalar	node	2.2×10^5	best	1 gene, its own label x condition
Total (68 datasets)			**4,561,865**				6.4×10^9		

2 The short list for a gene-sequence encoding benchmark ✗

2.1 What is being benchmarked ✗

GenSLM (Zvyagin et al. 2023, mirror key `zvyaginGenSLMsGenomescaleLanguage2023`) tokenizes a gene as codons and was pretrained on more than 110 million prokaryotic gene sequences from BV-BRC. `torchcell/datasets/genslm.py` embeds each gene's spliced coding sequence, in frame from the start codon and truncated to the first 2,048 codons when longer, and stores the masked mean of the last hidden layer: 512 values per gene for the 25M checkpoint and 1,840 for the 250M one, both of which are on GilaHyper with a sha256 manifest. The stores built so far are for *P. putida* KT2440; the *E. coli* K-12 MG1655 and BW25113 genomes expose a coding sequence through the same protocol, so their stores are a build away, not a code change.

The question a benchmark asks of such an encoding is narrow. Take a dataset whose label attaches to a gene, hold out a set of genes, fit a small head on the vectors of the remaining genes, and score the held-out ones. That score has meaning only against two controls at the same width: a one-hot or random vector per gene, which carries gene identity and nothing else, and a codon-frequency vector, which carries the one feature of a coding sequence a language model is sure to have learned. A codon model earns its place when it beats both; it is doing something different from either when it beats one and not the other.

2.2 The selection rule ✗

Three conditions, applied in order to every row of Table 2.

The label sits on one gene or one gene pair. The sequence-only fit tier must be *best*, or *good* only because the gene is crossed with a condition, since the condition is a second input the gene encoding does not have to carry. This removes every whole-strain readout: the titers, the growth rates of an engineered strain, the multi-omics panels of a wild type where the label is a profile of a strain rather than of a gene. It also removes the 6 stores whose label sits on a guide RNA rather than on the gene it targets (Sec. 3).

Enough labeled genes for a held-out split. A benchmark scored on a few dozen held-out genes cannot separate two encodings. Every core pick carries between 2,359 and 4,181 genes with a label, read from a full scan of the store, so a ten percent test fold holds hundreds of genes. The count that matters is genes, not instances: a proteome of 21 conditions is 21 instances and 2,359 labeled genes, and it is the latter that is split.

One primary per task family. A short list is one dataset per kind of question, with the datasets that pose the same question named beside it rather than listed on their own. The families are a binary call per gene, a quantitative fitness per gene and condition, a molecular level of the gene's own product, and a pairwise label.

2.3 The picks ✗

Table 3 is the list. The four core rows cover the four families and should be run first; the four second-tier rows add a translation-level, a transcript-level, an imaging and a knockdown task once the core four have been scored.

Essentiality, Goodall 2018. One TraDIS call per gene in each of 2 media, 4,181 genes and 8,309 calls. It is the simplest gene-level label in the host, and the one with three independent calls to agree with, from the Keio construction outcome (Baba 2006), genetic footprinting (Gerdes 2003) and the TnSeq no-insertion list (Price 2018). Genes are keyed by BW25113 locus tag, released as gene symbol.

Chemical genomics, Nichols 2011. A colony-size S-score for every Keio deletion in 324 conditions: 3,949 genes, 1,286,962 scores. It is the widest condition panel on one gene set in the host, so a held-out gene's whole profile is the target. The RB-TnSeq compendium of Price 2018 poses the same task with a sequencing readout on 104 conditions and is the replication arm. Genes are keyed by BW25113 locus tag, released as ECK accession.

Proteome, Schmidt 2016. Absolute copies per cell for 2,359 proteins in 21 growth conditions, 48,335 values, on an unperturbed wild type. The input is the coding sequence of the protein being measured, so this is the row where a codon model is expected, untested here, to separate most from a one-hot control. Mori 2021 measures the same quantity on a different platform and seven samples; Caglar 2017 is the REL606 panel. Genes are keyed by gene symbol (readout key).

Gene interaction, Babu 2014. A signed S score per deletion pair, 42,705 pairs over 3,880 genes. It is the only pairwise label at scale in the host and asks whether two gene vectors compose, which no single-gene task does. Two splits are worth scoring: both pair members unseen, and one member unseen. Butland 2008 is the unfiltered matrix

of the same screens. Genes are keyed by MG1655 b-number, released as Keio JW number; b-number with a Keio sub-number; gene symbol.

Second tier. Li 2014 gives an absolute protein synthesis rate per gene from ribosome profiling in 3 media (3,410 genes), the phenotype closest to what a codon model reads. PRECISE-1K (Lamoureux 2023) gives a TPM per gene in 119 distinct environments over 1,035 libraries (4,257 genes), the transcript-level counterpart of the proteome pick. Campos 2018 gives a vector of cell-shape and cell-cycle features per Keio deletion (3,664 genes), the one imaging phenotype in the host. Wang 2018 gives a gene-level CRISPRi fitness per gene and screen (4,223 genes, 20,158 values), the knockdown counterpart of the deletion screens, and the one that reaches essential genes.

Table 3: The short list for a gene-sequence encoding benchmark. *Genes*, *Labels*, *Env* and *Gene identifier* are read from a full scan of each store: distinct gene ids carrying a label, gene-attached numeric or boolean values, distinct stored environments, and the namespace the store keys genes by. *Task*, *Label*, *Split*, *Same task* and *Why* are the pick list in the generating script.

#	Dataset	Task	Label	Genes	Labels	Env	Gene identifier	Split	Same task	Why
Core										
1	GeneEssentiality Goodall2018	binary classification	essential or not, one TraDIS call per gene and medium	4,181	8,309	2	BW25113 locus tag, released as gene symbol	held-out genes	GeneEssentiality Baba2006, Gene EssentialityGerdes2003, GeneEssentiality Price2018Ecoli	The simplest gene-level label in the host, and the one with three independent calls to agree with (Keio construction, genetic footprinting, TnSeq). A classifier on one vector per gene is the whole model.
2	EnvChemgen Nichols2011	regression, gene x condition	a colony-size S-score per (deletion, condition)	3,949	1,286,962	324	BW25113 locus tag, released as ECK accession	held-out genes, every condition	RbTnseqPrice2018 Ecoli, EnvChemgen Shiver2016, Carbon SourceTong2020, Env ChemgenGirgis2009	The widest condition panel on one gene set. A held-out gene's whole profile is the target, and the condition enters as a second input the gene encoding does not have to carry.
3	Proteome Schmidt2016	regression, protein x condition	absolute copies per cell of one protein in one growth condition	2,359	48,335	21	gene symbol (readout key)	held-out genes, every condition	ProteomeMori2021, ProteomeCaglar2017, ProteomeIshii2007	A molecular level of the gene's own product, unperturbed, across conditions. Expected (untested) to be where a codon model separates from a one-hot control, because the input is the coding sequence whose product is measured.
4	GeneInteraction Babu2014	regression, gene pair	a signed colony-size S score per (query, array) deletion pair	3,880	42,705	1	MG1655 b-number, released as Keio JW number; b-number with a Keio sub-number; gene symbol	held-out genes (both pair members unseen), and one member unseen	GeneInteraction Butland2008, Crispri CrossRachwalski2024	The only pairwise label at scale in the host. It asks whether two gene vectors compose, which no single-gene task does.
Second tier										
5	ProteinSynthesis RateLi2014	regression, gene x medium	absolute protein synthesis rate per gene from ribosome profiling	3,410	8,644	3	gene symbol (readout key)	held-out genes	ProteinTurnover Gupta2024	The phenotype closest to what a codon model reads: translation of the coding sequence itself, in three media.
6	Rnaseq Lamoureux2023	regression, gene x library	TPM of one gene in one RNA-seq library	4,257	4,405,995	119	b-number (readout key)	held-out genes, every library	RnaseqPublicK12 Lamoureux2023, RnaseqCaglar2017	The transcript-level counterpart of the proteome pick, over a thousand libraries, so the per-gene target is a long expression profile.
7	Morphology Campos2018	multi-output regression	a vector of cell-shape and cell-cycle features per deletion	3,664	53,508	1	BW25113 locus tag	held-out genes	GrowthRate Campos2018	The one imaging phenotype in the host: a different label family from growth and abundance, on the same Keio gene set.
8	CrispriGeneFitness Wang2018	regression, gene x screen	gene-level CRISPRi fitness per (gene, screen)	4,223	20,158	5	MG1655 b-number	held-out genes	CrispriKnockdown Cui2018, Crispri ChemgenChoe2025, MismatchCrispri FitnessHawkins2020	The knockdown counterpart of the deletion screens, already aggregated to the gene by its authors, so it covers essential genes the deletion collections cannot.

3 What a score on these does not show ✗

Gene identifiers are the released ones. Each store keys its genes as its release does, and the *Gene identifier* column of Table 3 is measured from the records: b-numbers in one store, BW25113 locus tags or ECK accessions in another, bare symbols in a third. Resolving those to the coding sequences of the pinned assembly is a modeling-time step, by design, and its coverage must be measured before any score is read, because a gene that fails to resolve is silently absent from both the training and the test fold. The ECK synonym is the published one-to-one key between the two K-12 strains and is the route for the Keio-based rows.

Held-out genes can leak through paralogs. A paralog of a held-out gene in the training fold gives a sequence model a near copy of the test input. A split that clusters genes by sequence similarity before holding out clusters removes that route; a random split does not, and a random-split score is an upper bound.

The pretraining corpus is not disjoint from the test set. GenSLM was pretrained on gene sequences from BV-BRC, a database that holds *E. coli* K-12 assemblies, so the genes being scored are expected, unverified for any one gene, to be in its training data. The vector can then carry gene identity rather than a property of the sequence, which is exactly what the one-hot control measures: an encoding that scores no better than a one-hot vector of the same width has memorized, not generalized.

Guide-level stores are excluded on purpose. The 6 CRISPRi stores that score one guide at a time are the largest *E. coli* rows, and the gene each guide targets does have a coding sequence. The label, though, depends on the 20-nt protospacer, its position in the gene and its knockdown strength, none of which the gene's vector sees. Aggregating guides to their gene is what Wang 2018 released, and that is the row on the list.

A tier is not a result. The sequence-only fit tier and the pick list say what a sequence-only model could be given. No encoding has been scored on any of these rows here; every expectation in Sec. 2 is labeled as one.